1 Problem formulation

Let X_i be i.i.d. from the density f in $\mathbb{R}^d$. A discrete (anisotropic) Gaussian mixture model (GMM) assumes $X \sim \sum_k \mu_k \mathcal{N}(\boldsymbol{m}_k, V_k^2)$ yielding

$$f(\boldsymbol{x}) = \sum_{k \le K} \mu_k \varphi_{\boldsymbol{m}_k, V_k^2}(\boldsymbol{x}), \qquad \mu_k \ge 0, \qquad \sum_{k \le K} \mu_k = 1. \tag{1}$$

Here, $\varphi(\boldsymbol{x}) = \mathtt{C}_d \exp(-\|\boldsymbol{x}\|^2/2)$ for $\mathtt{C}_d = (2\pi)^{-d/2}$. An isotropic GMM corresponds to the case with $V_k^2 = \sigma_k^2 I_d$, and it can be coded by the parameter vector $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, \mu_k), k \le K\}$:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{\theta}) = \sum_{k \le K} \mu_k \sigma_k^{-d} \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}_k}{\sigma_k}\right). \tag{2}$$

Consider the problem of recovering the parameter $\boldsymbol{\theta}$ from the data.

The proposed procedure is based on method of moments. Given a family of test functions $(\omega_j(\boldsymbol{x}), j \le J)$, define the observables

$$Z_j \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \omega_j(X_i), \qquad j \le J.$$

One example is given by a collection of functions of the form

$$\omega_j(\boldsymbol{x}) = \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}\left(\frac{\boldsymbol{x} - \mathbb{x}_j}{s_j}\right). \tag{3}$$

for a kernel $\mathcal{K}$, a fixed set of points $\mathbb{x}_j \in \mathbb{R}^d$, directions $\boldsymbol{u}_j \in \mathbb{R}^d$, and scalings s_j, $j = 1, \dots, J$. Denote

$$\begin{aligned}
\psi_j(\boldsymbol{m}, \sigma) &\stackrel{\text{def}}{=} \sigma^{-d} \int \omega_j(\boldsymbol{x}) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x} \\
&= \int \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}\left(\frac{\sigma \boldsymbol{x}}{s_j}\right) \varphi(\boldsymbol{x} - \boldsymbol{m} + \mathbb{x}_j) d\boldsymbol{x}.
\end{aligned}$$

For a special kernel $\mathcal{K}(\boldsymbol{x})$, the quantity $\psi_j(\boldsymbol{m}, \sigma)$ can be computed analytically.

Lemma 1.1. *With* $\mathcal{K}(\boldsymbol{x}) = \exp(-\|\boldsymbol{x}\|^2/2)$

$$\begin{aligned}
\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma) &\stackrel{\text{def}}{=} \frac{1}{\sigma^d} \int \mathcal{K}\left(\frac{\boldsymbol{x} - \mathbb{x}}{s}\right) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x} \\
&= \left(1 + \frac{\sigma^2}{s^2}\right)^{-d/2} \exp\left\{-\frac{\|\boldsymbol{m} - \mathbb{x}\|^2}{2(\sigma^2 + s^2)}\right\}.
\end{aligned} \tag{4}$$

Furthermore,

$$\psi(\mathbb{x}, \boldsymbol{u}, s; \boldsymbol{m}, \sigma) \stackrel{\text{def}}{=} \frac{1}{\sigma^d} \int \langle \boldsymbol{x} - \mathbb{x}, \boldsymbol{u} \rangle \mathcal{K}\left(\frac{\boldsymbol{x} - \mathbb{x}}{s}\right) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x}$$
$$= \frac{\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma)}{\sigma^2/s^2 + 1} \langle \boldsymbol{m} - \mathbb{x}, \boldsymbol{u} \rangle. \tag{5}$$

Proof. By definition, it holds with $\boldsymbol{\delta} \stackrel{\text{def}}{=} (\boldsymbol{m} - \mathbb{x})/\sigma$

$$\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma) = \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2}\right) \varphi(\boldsymbol{x} - \boldsymbol{\delta}) d\boldsymbol{x}$$
$$= \mathtt{C}_d \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x}.$$

With $\rho \stackrel{\text{def}}{=} s^2/(\sigma^2 + s^2)$, it holds

$$\frac{\sigma^2}{s^2} \|\boldsymbol{x}\|^2 + \|\boldsymbol{x} - \boldsymbol{\delta}\|^2 = \frac{\sigma^2 + s^2}{s^2} \|\boldsymbol{x}\|^2 - 2\langle \boldsymbol{\delta}, \boldsymbol{x} \rangle + \|\boldsymbol{\delta}\|^2$$
$$= \frac{\sigma^2 + s^2}{s^2} \|\boldsymbol{x}\|^2 - 2\langle \boldsymbol{\delta}, \boldsymbol{x} \rangle + \frac{s^2}{\sigma^2 + s^2} \|\boldsymbol{\delta}\|^2 + \frac{\sigma^2}{\sigma^2 + s^2} \|\boldsymbol{\delta}\|^2$$
$$= \rho^{-1} \|\boldsymbol{x} - \rho\boldsymbol{\delta}\|^2 + (1 - \rho) \|\boldsymbol{\delta}\|^2 .$$

Therefore,

$$\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma) = \mathtt{C}_d \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x}$$
$$= \exp\left(-\frac{(1 - \rho) \|\boldsymbol{\delta}\|^2}{2}\right) \mathtt{C}_d \int \exp\left(-\frac{\|\boldsymbol{x} - \rho\boldsymbol{\delta}\|^2}{2\rho}\right) d\boldsymbol{x} \tag{6}$$
$$= \exp\left(-\frac{(1 - \rho) \|\boldsymbol{\delta}\|^2}{2}\right) \rho^{d/2},$$

and (4) follows by

$$(1 - \rho) \|\boldsymbol{\delta}\|^2 = \frac{\|\boldsymbol{m} - \mathbb{x}\|^2}{\sigma^2 + s^2}.$$

Further,

$$\int (\boldsymbol{x} - \mathbb{x}) \exp\left(-\frac{\|\boldsymbol{x} - \mathbb{x}\|^2}{2s^2}\right) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x}$$
$$= \sigma \mathtt{C}_d \int \boldsymbol{x} \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x},$$

and $\int \boldsymbol{x} e^{-\|\boldsymbol{x}\|^2/2} d\boldsymbol{x} = 0$ yields

$$
\begin{aligned}
&\int \boldsymbol{x} \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x}-\boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x} \\
&= \int \boldsymbol{x} \exp\left(-\frac{\|\boldsymbol{x}-\rho\boldsymbol{\delta}\|^2}{2\rho} - \frac{(1-\rho)\|\boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x} \\
&= \exp\left(-\frac{(1-\rho)\|\boldsymbol{\delta}\|^2}{2}\right) \rho\boldsymbol{\delta} \int \exp\left(-\frac{\|\boldsymbol{x}-\rho\boldsymbol{\delta}\|^2}{2\rho}\right) d\boldsymbol{x} \\
&= \rho\boldsymbol{\delta} \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x}-\boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x}.
\end{aligned}
$$

This and (6) complete the proof. $\square$

Under (2), it holds for $\omega_j(\boldsymbol{x})$ from (3) and $\psi(\mathbb{x}, \boldsymbol{u}, s; \boldsymbol{m}, \sigma)$ from (5)

$$
\int \omega_j(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x} = \sum_{k \le K} \mu_k \psi(\mathbb{x}_j, \boldsymbol{u}_j, s_j; \boldsymbol{m}_k, \sigma_k). \tag{7}
$$

1.1 Objective function

The idea of the approach is to approximate the true moments $\mathbb{E} Z_j = \int \omega_j(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x}$ by the sums from (7) with some parameters $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, \mu_k), k \le K\}$. In particular, if the model assumption (2) is correct with the true parameters $\boldsymbol{\theta}^* = \{(\boldsymbol{m}_k^*, \sigma_k^*, \mu_k^*)\}$, then the proposed procedure recovers this parameter vector with a small estimation error.

Introduce the objective function

$$
\mathcal{L}(\boldsymbol{\theta}) \stackrel{\text{def}}{=} \frac{1}{2} \|\boldsymbol{Z} - \boldsymbol{\psi}(\boldsymbol{\theta})\|^2,
$$

where $\boldsymbol{\psi}(\boldsymbol{\theta})$ is a vector in $\mathbb{R}^q$ with the entries

$$
\psi_j(\boldsymbol{\theta}) = \sum_{k \le K} \mu_k \psi_j(\boldsymbol{m}_k, \sigma_k).
$$

If $\omega_j(\boldsymbol{x}) = \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}((\boldsymbol{x} - \mathbb{x}_j)/s_j)$ as in (3) then by (4)

$$
\psi_j(\boldsymbol{\theta}) = \sum_{k \le K} \mu_k \psi(\mathbb{x}_j, \boldsymbol{u}_j, s_j; \boldsymbol{m}_k, \sigma_k).
$$

Note that $\boldsymbol{\mu} = (\mu_k)$ is a point in a simplex in $\mathbb{R}^K$. It is often useful to code such a point using

$\alpha_k = \overline{\alpha} + \log \mu_k$. This leads to a representation

$$\psi_j(\boldsymbol{\theta}) = \frac{1}{\sum_{k \le K} e^{\alpha_k}} \sum_{k \le K} e^{\alpha_k} \psi(\mathbb{x}_j, \boldsymbol{u}_j, s_j; \boldsymbol{m}_k, \sigma_k).$$

A benefit of this form is that the vector $\boldsymbol{\alpha} = (\alpha_k)$ can be any point in $\mathbb{R}^K$.

Unfortunately, we face an identifiability issue if the value K exceeds the true number of Gaussian components. To overcome this problem, introduce an entropy penalty

$$\operatorname{pen}(\boldsymbol{\mu}) \stackrel{\text{def}}{=} \lambda \operatorname{Ent}(\boldsymbol{\mu}), \qquad \operatorname{Ent}(\boldsymbol{\mu}) \stackrel{\text{def}}{=} -\sum_{k \le K} \mu_k \log \mu_k,$$

$$\operatorname{pen}(\boldsymbol{\alpha}) \stackrel{\text{def}}{=} -\lambda \sum_{k \le K} e^{\alpha_k - \overline{\alpha}} (\alpha_k - \overline{\alpha}), \qquad \overline{\alpha} \stackrel{\text{def}}{=} \log \sum_{k \le K} e^{\alpha_k},$$

which strongly pushes to zero components with small weights μ_k. Such a penalty ensures identifiability in the number of components. The identifiability problem is still unresolved because the components can be exchanged. One possibility to make it fixed is by using an additional penalty $\operatorname{pen}(\boldsymbol{m})$ based on the distance of each mean $\boldsymbol{m}_k$ to a preliminary guess $\boldsymbol{m}_{0,k}$):

$$\operatorname{pen}(\boldsymbol{m}) \stackrel{\text{def}}{=} \frac{\lambda_m}{2} \sum_{k \le K} \|\boldsymbol{m}_k - \boldsymbol{m}_{0,k}\|^2 = \frac{\lambda_m}{2} \|\boldsymbol{m} - \boldsymbol{m}_0\|^2 .$$

Altogether leads to the following approach: for $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, \alpha_k)\}$

$$\begin{aligned} \widetilde{\boldsymbol{\theta}} &= \operatorname*{arginf}_{\boldsymbol{\theta}} \{\mathcal{L}(\boldsymbol{\theta}) + \operatorname{pen}(\boldsymbol{m}) + \operatorname{pen}(\boldsymbol{\mu})\} \\ &= \operatorname*{arginf}_{\boldsymbol{\theta}} \left\{ \frac{1}{2} \|\boldsymbol{Z} - \boldsymbol{\psi}(\boldsymbol{\theta})\|^2 + \operatorname{pen}(\boldsymbol{\alpha}) + \operatorname{pen}(\boldsymbol{m}) \right\}. \end{aligned} \tag{8}$$

The structural moments $\boldsymbol{\psi}(\boldsymbol{\theta})$ creates some difficulties for the analysis, however, it is deterministic and smooth in the scope of arguments.

1.2 Dimension-free formulation

The ambient dimension d shows up only in the scaling $(1 + \sigma^2/s^2)^{-d/2}$ of (4). This factor only depends on the variance σ of the Gaussian component and the parameter s. We aim at eliminating this dependence by fusing it with the factor μ_k. This leads to another representation of $\mathbb{E}Z_j$. Consider first the case when the value s does not change over the points $\mathbb{x}_j$. Then the scaling only depends on σ_k. With $\omega_j(\boldsymbol{x}) = \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}((\boldsymbol{x} - \mathbb{x}_j)/s)$, it holds for $f(\boldsymbol{x}, \boldsymbol{\theta}) = \sum_{k \le K} \mu_k \varphi_{\boldsymbol{m}_k, \sigma_k}(\boldsymbol{x})$

$$\begin{aligned}\int \omega_j(\boldsymbol{x}) f(\boldsymbol{x}, \boldsymbol{\theta}) d\boldsymbol{x} &= \sum_{k \le K} \mu_k \left(1 + \frac{\sigma_k^2}{s^2}\right)^{-d/2} \exp\left\{-\frac{\|\boldsymbol{m}_k - \mathbb{x}_j\|^2}{2(\sigma_k^2 + s^2)}\right\} \frac{1}{\sigma_k^2/s^2 + 1} \langle \boldsymbol{m}_k - \mathbb{x}_j, \boldsymbol{u}_j \rangle \\ &= \sum_{k \le K} e^{c_k} \exp\left\{-\frac{\|\boldsymbol{m}_k - \mathbb{x}_j\|^2}{2(\sigma_k^2 + s^2)}\right\} \frac{1}{\sigma_k^2/s^2 + 1} \langle \boldsymbol{m}_k - \mathbb{x}_j, \boldsymbol{u}_j \rangle,\end{aligned}$$

where the values c_k replace α_k:

$$e^{c_k} \stackrel{\text{def}}{=} \mu_k \left(1 + \frac{\sigma_k^2}{s^2}\right)^{-d/2} = e^{\alpha_k} \left(1 + \frac{\sigma_k^2}{s^2}\right)^{-d/2}. \tag{9}$$

With $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, c_k), k \le K\}$, we apply (8) and for $j \le J$

$$\psi_j(\boldsymbol{\theta}) = \sum_{k \le K} e^{c_k} \exp\left\{-\frac{\|\boldsymbol{m}_k - \mathbb{x}_j\|^2}{2(\sigma_k^2 + s^2)}\right\} \frac{1}{\sigma_k^2/s^2 + 1} \langle \boldsymbol{m}_k - \mathbb{x}_j, \boldsymbol{u}_j \rangle.$$

2 Second-order moments

Zeroth- and first-order moments are well suited to estimating the overall mass and the locations of the component centers, respectively, but are less informative about the covariance. For instance, suppose that the test center $\mathbb{x}$ is close to the estimated mean m of a component, so that $m - \mathbb{x} \approx 0$. Then

$$\psi(\mathbb{x}, \boldsymbol{u}, s; \boldsymbol{m}, \sigma) = \frac{\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma)}{\sigma^2/s^2 + 1} \langle \boldsymbol{m} - \mathbb{x}, \boldsymbol{u} \rangle \approx 0. \tag{10}$$

Thus, this equation provides little information about σ.

To address this issue, we introduce second-order moments. We consider two types of second-order moments: isotropic and anisotropic, corresponding to diagonal and off-diagonal terms of the covariance matrix, respectively. By isotropic second-order moments, we mean

$$\omega^2(x) = \langle x - \mathbb{x}, u \rangle^2 \, \mathcal{K}\left(\frac{x - \mathbb{x}}{s}\right) \tag{11}$$

and by anisotropic second-order moments, we mean

$$\omega^{1,1}(x) = \langle x - \mathbb{x}, u \rangle \langle x - \mathbb{x}, v \rangle \, \mathcal{K}\left(\frac{x - \mathbb{x}}{s}\right) \tag{12}$$

For our model, we therefore need to define the following quantities and derive explicit expressions for them:

$$\psi^2(m,\sigma) \stackrel{\text{def}}{=} \sigma^{-d} \int \omega^2(x)\varphi\left(\frac{x-m}{\sigma}\right)\, dx \tag{13}$$

$$\psi^{1,1}(m,\sigma) \stackrel{\text{def}}{=} \sigma^{-d} \int \omega^{1,1}(x)\varphi\left(\frac{x-m}{\sigma}\right)\, dx \tag{14}$$

Lemma 2.1. *For* $\mathcal{K}(\boldsymbol{x}) = \exp(-\|\boldsymbol{x}\|^2/2)$, *set*

$$\rho \stackrel{\text{def}}{=} \frac{s^2}{\sigma^2+s^2} = \frac{1}{\sigma^2/s^2+1}.$$

Then

$$\begin{aligned}\psi^{1,1}(m,\sigma) &\stackrel{\text{def}}{=} \sigma^{-d}\int \langle x-\mathbb{x},u\rangle\langle x-\mathbb{x},v\rangle\, \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right)\varphi\left(\frac{x-m}{\sigma}\right)\, dx \\ &= \psi(\mathbb{x},s;m,\sigma)\left\{\rho^2\langle m-\mathbb{x},u\rangle\langle m-\mathbb{x},v\rangle + \sigma^2\rho\langle u,v\rangle\right\}.\end{aligned}$$

Furthermore,

$$\begin{aligned}\psi^{2}(m,\sigma) &\stackrel{\text{def}}{=} \sigma^{-d}\int \langle x-\mathbb{x},u\rangle^2\, \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right)\varphi\left(\frac{x-m}{\sigma}\right)\, dx \\ &= \psi(\mathbb{x},s;m,\sigma)\left\{\rho^2\langle m-\mathbb{x},u\rangle^2 + \sigma^2\rho\|u\|^2\right\}.\end{aligned}$$

Proof. Put $\delta \stackrel{\text{def}}{=} (m-\mathbb{x})/\sigma$. After the change of variables $y=(x-\mathbb{x})/\sigma$, we obtain

$$\psi^{1,1}(m,\sigma) = \sigma^2\mathtt{C}_d\int \langle y,u\rangle\langle y,v\rangle \exp\left(-\frac{\sigma^2\|y\|^2}{2s^2} - \frac{\|y-\delta\|^2}{2}\right) dy.$$

As in the proof of (4), the definition of ρ yields

$$\frac{\sigma^2}{s^2}\|y\|^2 + \|y-\delta\|^2 = \rho^{-1}\|y-\rho\delta\|^2 + (1-\rho)\|\delta\|^2.$$

For every $u,v\in\mathbb{R}^d$, symmetry and the second moments of the centered isotropic Gaussian distribution give

$$\int \langle z,u\rangle \exp\left(-\frac{\|z\|^2}{2\rho}\right) dz = 0$$

and

$$\int \langle z,u\rangle\langle z,v\rangle \exp\left(-\frac{\|z\|^2}{2\rho}\right) dz = \rho\langle u,v\rangle\int \exp\left(-\frac{\|z\|^2}{2\rho}\right) dz.$$

Consequently, the change of variables $z = y - \rho\delta$ gives

$$\begin{aligned}\psi^{1,1}(m,\sigma) &= \sigma^2 \mathtt{C}_d \exp\left(-\frac{(1-\rho)\|\delta\|^2}{2}\right) \int \langle z+\rho\delta, u\rangle\langle z+\rho\delta, v\rangle \exp\left(-\frac{\|z\|^2}{2\rho}\right) dz \\ &= \sigma^2 \mathtt{C}_d \exp\left(-\frac{(1-\rho)\|\delta\|^2}{2}\right) \left\{\rho\langle u,v\rangle + \rho^2\langle\delta,u\rangle\langle\delta,v\rangle\right\} \int \exp\left(-\frac{\|z\|^2}{2\rho}\right) dz \\ &= \psi(\mathbb{x}, s; m, \sigma) \left\{\sigma^2\rho\langle u,v\rangle + \rho^2\langle m-\mathbb{x}, u\rangle\langle m-\mathbb{x}, v\rangle\right\}.\end{aligned}$$

Here, the last identity follows from (6) and $\sigma\delta = m - \mathbb{x}$. The expression for $\psi^2(m,\sigma)$ follows by setting $v = u$. □

3 Radial second-order moments

The second-order moments introduced in the previous section depend on a direction u. This directional dependence is necessary for recovering a general anisotropic covariance matrix, but it may be unnecessarily noisy when the components are isotropic and the goal is only to estimate their scalar variances. A natural alternative is to average the directional second-order moments over all directions.

For a test center $\mathbb{x} \in \mathbb{R}^d$ and a scale $s > 0$, define the radial second-order test function by

$$\omega^{\mathrm{rad}}(x) \stackrel{\mathrm{def}}{=} \|x - \mathbb{x}\|^2 \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{15}$$

It is also convenient to use its dimensionless, direction-averaged normalization

$$\overline{\omega}^{\mathrm{rad}}(x) \stackrel{\mathrm{def}}{=} \frac{\|x - \mathbb{x}\|^2}{ds^2} \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{16}$$

The normalization does not change the information carried by the moment. It only puts its magnitude on the same scale as that of one normalized directional second-order moment.

For the Gaussian kernel

$$\mathcal{K}(z) = \exp\left(-\frac{\|z\|^2}{2}\right),$$

both functions vanish at $x = \mathbb{x}$. Their pointwise mass is concentrated around a sphere: as a function of $r = \|x - \mathbb{x}\|$, the quantity $r^2 \exp(-r^2/(2s^2))$ is maximized at $r = \sqrt{2}s$. Thus, unlike the zeroth-order Gaussian test function, the radial second-order test function does not put its largest weight at the test center itself. Notice that after including the radial volume element $r^{d-1}dr$, the corresponding radial mass is instead maximized at $r = \sqrt{d+1}s$.

Let

$$\psi^{\mathrm{rad}}(m,\sigma) \stackrel{\mathrm{def}}{=} \sigma^{-d} \int \omega^{\mathrm{rad}}(x)\varphi\left(\frac{x-m}{\sigma}\right) dx,$$
$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) \stackrel{\mathrm{def}}{=} \sigma^{-d} \int \overline{\omega}^{\mathrm{rad}}(x)\varphi\left(\frac{x-m}{\sigma}\right) dx.$$

Lemma 3.1. *For $\mathcal{K}(z) = \exp(-\|z\|^2/2)$, set*

$$\rho \stackrel{\mathrm{def}}{=} \frac{s^2}{\sigma^2+s^2}.$$

Then

$$\psi^{\mathrm{rad}}(m,\sigma) = \psi(\mathbb{x}, s; m, \sigma)\left\{\rho^2\|m-\mathbb{x}\|^2 + d\sigma^2\rho\right\}. \tag{17}$$

Consequently,

$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) = \psi(\mathbb{x}, s; m, \sigma)\left\{\frac{s^2\|m-\mathbb{x}\|^2}{d(s^2+\sigma^2)^2} + \frac{\sigma^2}{s^2+\sigma^2}\right\}. \tag{18}$$

Proof. Let $e_1, \ldots, e_d$ be an orthonormal basis of $\mathbb{R}^d$. Since

$$\|x-\mathbb{x}\|^2 = \sum_{\ell=1}^{d}\langle x-\mathbb{x}, e_\ell\rangle^2,$$

the radial moment is the sum of the directional second-order moments over this basis. Applying the formula from the previous section with $u = e_\ell$ and summing over ℓ gives

$$\begin{aligned}\psi^{\mathrm{rad}}(m,\sigma) &= \psi(\mathbb{x}, s; m, \sigma)\sum_{\ell=1}^{d}\left\{\rho^2\langle m-\mathbb{x}, e_\ell\rangle^2 + \sigma^2\rho\|e_\ell\|^2\right\}\\ &= \psi(\mathbb{x}, s; m, \sigma)\left\{\rho^2\|m-\mathbb{x}\|^2 + d\sigma^2\rho\right\}.\end{aligned}$$

This proves (17). Dividing it by ds^2 and substituting $\rho = s^2/(s^2+\sigma^2)$ gives (18). □

The radial construction can also be viewed as an exact average of directional moments. Indeed,

$$\overline{\omega}^{\mathrm{rad}}(x) = \frac{1}{d}\sum_{\ell=1}^{d}\frac{\langle x-\mathbb{x}, e_\ell\rangle^2}{s^2}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{19}$$

Thus, for isotropic components, it aggregates the second-order information from all orthogonal directions without introducing an additional directional design. It does not, however, retain the directional information needed to recover a general anisotropic covariance matrix.

3.1 Relation to the scale derivative

For the Gaussian kernel, the radial second-order test function has a simple relation to the derivative of the zeroth-order test function with respect to its scale:

$$\boxed{\frac{\|x-\mathbb{x}\|^2}{s^2}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right) = s\frac{\partial}{\partial s}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right).} \tag{20}$$

Consequently, differentiation under the integral gives

$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) = \frac{s}{d}\frac{\partial}{\partial s}\psi(\mathbb{x},s;m,\sigma). \tag{21}$$

Hence a radial second-order moment measures the local change of the zeroth-order moment as the kernel scale varies. In particular, if zeroth-order moments are available on a dense grid of scales, part of the same information is already present through finite differences in s. The direct radial moment provides this scale derivative without numerical differencing.

3.2 Recovery of a component variance

Suppose first that a single component is considered and that the test center coincides with its mean, $\mathbb{x} = m$. Equations (4) and (18) yield

$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) = \psi(m,s;m,\sigma)\frac{\sigma^2}{s^2+\sigma^2}. \tag{22}$$

Therefore, with

$$q \stackrel{\mathrm{def}}{=} \frac{\overline{\psi}^{\mathrm{rad}}(m,\sigma)}{\psi(m,s;m,\sigma)},$$

we have $q = \sigma^2/(s^2+\sigma^2)$ and hence

$$\boxed{\sigma^2 = s^2\frac{q}{1-q} = s^2\frac{\overline{\psi}^{\mathrm{rad}}(m,\sigma)}{\psi(m,s;m,\sigma) - \overline{\psi}^{\mathrm{rad}}(m,\sigma)}.} \tag{23}$$

The component weight cancels from this ratio. The same identity can be used for one component of a mixture after explicitly subtracting the effects of all other components. Let the total zerothand radial moments evaluated at the center m_k be

$$G_k \stackrel{\text{def}}{=} \int \mathcal{K}\left(\frac{x-m_k}{s}\right) f(x)dx,$$
$$H_k \stackrel{\text{def}}{=} \int \frac{\|x-m_k\|^2}{ds^2}\mathcal{K}\left(\frac{x-m_k}{s}\right) f(x)dx.$$

For every $\ell \neq k$, define its contributions to these two moments by

$$G_{\ell\to k} \stackrel{\text{def}}{=} \mu_\ell \psi(m_k, s; m_\ell, \sigma_\ell), \tag{24}$$

$$H_{\ell\to k} \stackrel{\text{def}}{=} G_{\ell\to k}\left\{\frac{s^2\|m_\ell - m_k\|^2}{d(s^2+\sigma_\ell^2)^2} + \frac{\sigma_\ell^2}{s^2+\sigma_\ell^2}\right\}. \tag{25}$$

After removing these neighboring contributions, set

$$G_k^{\text{corr}} \stackrel{\text{def}}{=} G_k - \sum_{\ell\neq k} G_{\ell\to k}, \qquad H_k^{\text{corr}} \stackrel{\text{def}}{=} H_k - \sum_{\ell\neq k} H_{\ell\to k}.$$

The corrected quantities contain only the contribution of component k. Therefore,

$$\boxed{\sigma_k^2 = s^2\frac{H_k^{\text{corr}}}{G_k^{\text{corr}} - H_k^{\text{corr}}} = s^2\frac{H_k - \sum_{\ell\neq k} H_{\ell\to k}}{G_k - \sum_{\ell\neq k} G_{\ell\to k} - H_k + \sum_{\ell\neq k} H_{\ell\to k}}.} \tag{26}$$

Thus, while the parameters of the other components are held fixed, one can compute σ_k for a single component explicitly rather than optimize it. This gives a coordinate-wise profiled or alternating procedure: update the means and amplitudes, subtract the current neighboring contributions, and then update each variance using (26). For empirical moments, the corrected ratio can be restricted to an interval inside $(0,1)$ for numerical stability.

3.3 A compensated radial moment and uniform noise

The radial moment in (16) is nonnegative, so a zero-integral version must introduce a signed compensation. One possible choice is

$$\omega^{\text{rad},0}(x) \stackrel{\text{def}}{=} \left\{\frac{\|x-\mathbb{x}\|^2}{ds^2} - 1\right\}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right) = \varpi^{\text{rad}}(x) - \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{27}$$

For the Gaussian kernel, $\int_{\mathbb{R}^d} \omega^{\text{rad},0}(x)dx = 0$. Consequently, a locally uniform noise density gives an approximately zero response, which adds robustness to uniform contamination.

Numerical experiments confirm this robustness effect, but also show that the signed compensated moment makes the moment map and the resulting objective much more difficult to optimize. Thus, the increased robustness to uniform noise comes at the cost of a substantially harder optimization problem, and the quality of the underlying estimation procedure suffers.